Denoising-Driven RGB-T Tracking via Distillation and Dynamic Fusion

Fangmei Chen, Weixin Kong, Lifeng Wang, Fasheng Wang, Fuming Sun, Haojie Li

***Abstract*—RGB-Thermal (RGB-T) object tracking leverages cross-modal fusion to address tracking failures under extreme illumination and occlusion conditions, yet its performance remains constrained by fusion noise, modality-quality noise, and temporal bias. To tackle these challenges, we propose D^3-Track, a denoising-driven framework that systematically suppresses noise at the semantic, spatial, and temporal levels. Our method consists of three key components, each designed to address a distinct type of noise. At the semantic level, we propose a Hierarchical Cross-modal Knowledge-Alignment Distillation (HCKD) scheme to suppress fusion noise. HCKD replaces dense inference-time cross-modal interaction with training-time teacher-to-student guidance, thereby aligning RGB-T representations and suppressing unreliable responses before fusion. At the spatial level, we design a Token-wise Adaptive Gating & Recalibration (TAGR) module that achieves fine-grained control over modality-quality noise by dynamically evaluating and weighting the contribution of each token within the modalities, thereby effectively suppressing the negative impact of low-quality modal information. At the temporal level, we propose an Event-Driven Template Update (EDTU) strategy that establishes an update mechanism triggered by significant appearance changes, substantially enhancing the temporal robustness and template representation power during tracking. Our proposed modules are integrated into a ViT backbone for joint feature extraction, search-template matching, and cross-modal interaction. Extensive experiments on three popular RGB-T object tracking benchmarks demonstrate that our method delivers a new state-of-the-art (SOTA) performance. Code and results are released on https://github.com/KWX2025/D3Track.**

***Index Terms*—RGB-T tracking, Cross-modal fusion, Hierarchical cross-modal knowledge distillation, Reliability-aware fusion, Event-driven template update**

I. Introduction

RGB-Thermal tracking is an active research topic in multimodal multimedia processing community [1]. The key issue in RGB-T tracking is the adaptive fusion of RGB and thermal modalities [2]–[4], which aims to fully leverage the rich texture details of RGB modality and the reliable thermal radiation information from the TIR modality to cope with complex and variable tracking environments, such as sudden illumination changes, occlusion, background clutter, etc.

Manuscript received November 30, 2025; revised XX XX, XXXX. This work was supported in part by the National Natural Science Foundation of China under Grant 61972068, the Joint Funds of Liaoning Science and Technology Program under Grant 2023JH2/101800032, the Taishan Scholar Program of Shandong Province under Grant tstp20221128, and the Fundamental Research Funds for the Central Universities. (*Corresponding author: Fasheng Wang*)

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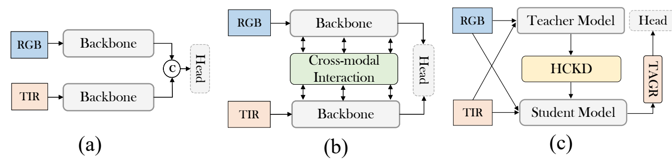

Fig. 1. RGB Thermal interaction paradigms: (a) Late fusion via post-encoder concatenation; (b) Dense global cross-modal attention; (c) D^3-Track (ours): HCKD teacher-student alignment with student-only inference (No cross-attention-based fusion).

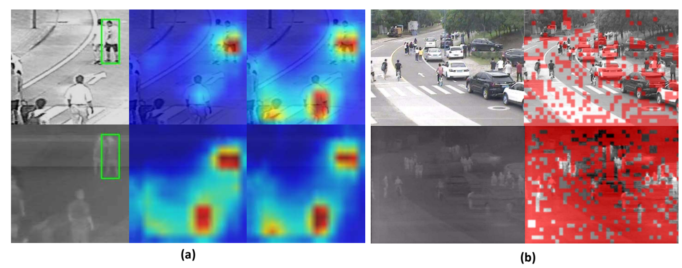

Fig. 2. Visualization of two degradation sources in RGB-T object tracking. (a) Spurious-response propagation in cross-attention-based fusion. The left column shows RGB and TIR frames with ground-truth boxes; the middle column shows unimodal responses (RGB-only / TIR-only), where unreliable modalities may contain spurious peaks; the right column shows bimodal responses from a cross-attention fusion tracker. (b) Modality-quality heterogeneity. PIQE-based block maps are overlaid on RGB/TIR frames, with red tiles indicating locally degraded regions (darker red = worse), highlighting spatially varying reliability within a frame.

Existing RGB-T tracking methods can be broadly classified into two paradigms based on the modality information interaction mechanisms. One is late fusion via post-encoder concatenation [5], [6]. As shown in Fig. 1(a), the RGB and TIR inputs are fed into independent encoder networks and then merged through a concatenation operation. However, such global and indiscriminate fusion methods introduce substantial background redundancy and degrade discrimination (e.g., Fig. 2(a)). The other paradigm employs cross-modal attention and related mechanisms [7]–[11] to perform global interaction, attempting to build dense RGB-TIR correspondences, as shown in Fig. 1(b). Although this paradigm can capture long-range dependencies, in complex scenes, the indiscriminate global interaction tends to propagate noise from unreliable modalities across the whole feature map, amplifying background interference and semantic mismatch. Furthermore, these methods typically lack explicit modeling of modality reliability, limiting their adaptability to the quality fluctuations under varying imaging conditions (see Fig. 2(b)). In summary, while the fusion of RGB and TIR modalities provides richer information, it comes with the inherent risk of propagating and amplifying noise.

In this work, "noise" does not refer to sensor-level corruption or explicit image degradation. Instead, it denotes task-induced unreliable information that may disturb RGB-T tracking, such as spurious background responses, modality-inconsistent activations, degraded modality cues, and outdated templates. From this perspective, denoising aims to suppress the propagation and accumulation of such unreliable information throughout the tracking pipeline. Accordingly, we group the main sources of unreliable information into three categories:

(1) **Fusion noise**: semantic mismatch and background clutter under indiscriminate global fusion.

(2) **Modality-quality noise**: condition-dependent imaging reliability disparities with spatial and temporal heterogeneity.

(3) **Temporal bias**: template staleness and mis-updates.

Fusion noise is mainly attributed to indiscriminate fusion (Fig. 1(a)) or cross-attention-based interaction (Fig. 1(b)), as these fusion strategies lack robustness inherently. In visually complex or degraded environments, such as crowded scenes, areas with strong reflections, or rainy and foggy conditions, such fusion strategies tend to propagate spurious responses from unreliable modalities across the entire feature map (see Fig. 2(a), compared to unimodal responses, cross-attention tends to propagate spurious activations from unreliable modalities, resulting in diffuse/background responses in both), thereby amplifying background clutter and leading to semantic mismatch.

Modality-quality noise stems from the condition-dependent imaging reliability of RGB and TIR modalities, which exhibits spatial and temporal heterogeneity. Spatially, quality disparities exist across different regions within a single frame (see Fig. 2(b)). For instance, RGB information may degrade in back-lit areas where TIR remains reliable, while TIR often suffers from thermal interference where RGB is cleaner. Temporally, the relative quality of the two modalities is dynamic and can reverse as imaging conditions change across consecutive frames.

Temporal bias arises from template anchoring and unstable update policies. To quantify this bias, we used the initial frame template as a reference and computed the Structural Similarity (SSIM$\in [0, 1]$, higher values indicate greater similarity) between it and subsequent frames. As shown in Fig. 8(a), the SSIM values exhibit a significant decline as the frame index increases. This monotonically decreasing trend clearly captures the gradual accumulation of temporal bias over the video sequence and demonstrates a strong correlation with the anchoring effect of the early template.

Building on these observations, we propose D^3-Track, a novel RGB-T tracking framework designed from a denoising perspective, **D**enoising-driven RGB-T Tracking via **D**istillation and **D**ynamic fusion. D^3-Track suppresses errors at three complementary levels—semantic, spatial, and temporal—through three dedicated components, as shown in Fig. 1(c).

At the semantic level, we propose Hierarchical Cross-modal Knowledge-Alignment Distillation (HCKD) to suppress unreliable response propagation during cross-modal representation learning.Instead of directly denoising already fused features, HCKD converts cross-modal information transfer into training-time teacher-to-student guidance. In this way, task-related discriminative knowledge from the opposite-modality teacher is internalized into the student parameters, thereby reducing the risk that abnormal responses from one modality are propagated to the other branch through dense cross-modal attention during inference. HCKD follows an 'align-then-transfer" paradigm, which first aligns teacher features to the student modality space and then transfers high-level discriminative cues in the aligned feature space, thereby effectively reducing the cross-modal propagation and layer-wise amplification of fusion noise. As shown in Fig. 3, the increased token-level similarity after alignment indicates improved RGB-T representation compatibility.

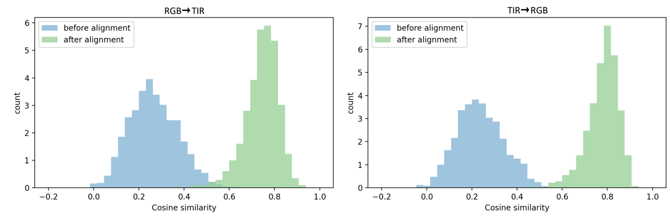

Fig. 3. Token-level cosine similarity histograms for cross-modal alignment. In each subfigure, blue bars denote similarities before alignment and green bars denote similarities after alignment.

At the spatial level, we design a Token-wise Adaptive Gating & Recalibration (TAGR) module and inject it before the tracking head. TAGR mitigates modality-quality noise via reliability-aware fusion, predicting two complementary and bounded token-wise weight maps per frame to form a proportionate blend that amplifies reliable cues and confines unreliable ones. The weighting mechanism prevents spatial/temporal spread of modality-quality noise while preserving discriminative structure.

At the temporal level, to mitigate temporal bias, we propose an Event-Driven Template Update (EDTU) strategy using a four-slot template pool. It enforces a stability-adaptivity balance via long-term anchoring, event-triggered updates, and sparse injection, transforming the memory into an evidence-based controller that suppresses error accumulation under various challenges.

The above-mentioned core components are seamlessly integrated into a Vision Transformer (ViT) backbone, which introduces negligible additional inference time. Extensive evaluations on multiple benchmarks demonstrate that D^3-Track achieves consistently superior performance, particularly in challenging conditions such as low illumination, strong reflections, and distractors.

The main contributions of this work are summarized as follows:

- We revisit RGB-T tracking from a denoising-driven perspective, explicitly analyzing how fusion noise, modality-quality noise, and temporal bias jointly degrade tracking robustness.
- We construct D^3-Track, a denoising-driven RGB-T tracking framework that suppresses errors at three complementary levels via three dedicated modules, HCKD for cross-modal semantic alignment, TAGR for token-wise

reliability-aware fusion, and EDTU for event-driven template management.
- Extensive experiments on three popular RGB-T object tracking benchmarks (RGBT234, LasHeR, and VTUAV) demonstrate that D^3-Track delivers new state-of-the-art performance.

II. Related work

A. RGB-T Tracking

Centering on the core challenge of noise suppression during fusion, recent RGB-T tracking research has advanced along two main directions.

Architectural integration and simplification. USTrack [8] integrates feature extraction, cross-modal interaction, and relation modeling into a unified single-stage ViT backbone, thereby reducing the noise accumulation and latency introduced by multi-stage fusion. Building on this idea, SUTrack [12] further extends to a unified large-model architecture that accommodates multiple multimodal tracking tasks, demonstrating the strong potential of unified designs in cross-modal scenarios.

In-depth and fine-grained control of interaction mechanisms. AINet [13] introduces a state-space formulation to realize hierarchical multimodal interaction and progressive fusion, enabling efficient inter-layer information exchange with linear complexity and effectively suppressing background-noise diffusion common in traditional fusion schemes. The TBSI series [14] proposes a template-mediated, bridge-style interaction path to mitigate the background-noise amplification caused by global explicit interactions.

Despite these advances, challenges persist under complex backgrounds and drastic modality-quality fluctuations. Although TBSI mitigates noise amplification via template-mediated interaction and IGF-style gated fusion, template-conditioned weighting may still suffer from (1) temporal lag—weights inferred from historical templates cannot be refreshed every frame and become misaligned when modality reliability shifts rapidly—and (2) unimodal evidence insufficiency, since single-modality template cues are under-determined for inferring bimodal reliability under spatially/temporally heterogeneous degradations (e.g., RGB saturation at night or thermal interference). Consequently, fusion may still introduce or propagate background interference, weakening multimodal complementarity. To address this, we predict token-wise complementary weights from current-frame joint RGB–TIR search evidence, avoiding template-lagged and unimodal template-conditioned weighting.

B. Cross-Modal Knowledge Distillation

Recently, knowledge distillation (KD) has been extended to cross-modal vision tasks, where knowledge is transferred from a teacher modality to a student modality. Recent studies [15]–[18] in multimodal tracking and representation learning suggest that the modality gap, covering distributional and semantic mismatches, is the principal obstacle that undermines CMKD.

C^{2}KD [15] empirically identifies two typical issues: modality imbalance and soft-label misalignment, which markedly weaken distillation gains. They address these with bidirectional distillation coupled with online filtering and sample discard (OFSD). For video representation, XKD [19] emphasizes domain alignment: within a self-supervised framework, it aligns cross-modal feature distributions and target spaces so that the student progressively approaches the teacher's representational capacity in the other modality, thereby mitigating distillation shift.

Focusing on multimodal tracking, CMD [20] systematizes cross-modal distillation for RGB-T and RGB-D tracking by transferring both common and specific features, narrowing the performance gap between a lightweight student and a strong teacher, and yielding consistent improvements across multiple benchmarks. From a style-centered perspective, CKD [16] first conducts style distillation to bring the statistical styles of RGB and TIR closer, followed by content distillation to preserve semantics. This approach substantially reduces the modality gap without additional inference overhead, offering a pragmatic alignment route for tracking tasks.

Although HCKD is related to recent RGB-T distillation methods such as CKD and CMDTrack, it differs from them in terms of objective, teacher–student topology, and distillation target. CKD mainly focuses on reducing the modality gap between RGB and TIR by style distillation between student branches, while using same-modality teachers to preserve content knowledge. CMDTrack is primarily designed for efficient RGB-T tracking, where a strong multimodal teacher transfers knowledge to a compact student through different levels of feature and prediction distillation. In contrast, HCKD does not aim at model compression or direct imitation of fused multimodal representations. Instead, it constructs a bidirectional heterogeneous cross-supervision scheme, where the RGB teacher supervises the TIR student and the TIR teacher supervises the RGB student. This design transfers opposite-modality discriminative cues before fusion, improving the representation compatibility and target discrimination of both student branches.

III. Proposed Method

A. Overview

The structure of the proposed D^3-Track is shown in Fig. 4. During training, it adopts a dual teacher – dual student design with four parameter-decoupled Vision Transformer (ViT) backbones: for each modality (RGB and TIR), one teacher and one student are instantiated. Supervision is jointly provided by a multimodal tracking head and two unimodal teacher heads. At inference, the two teachers and the two unimodal heads are removed, retaining only the two student branches, the TAGR module, and the multimodal tracking head, yielding a lightweight inference architecture.

Specifically, on the input side, we adopt a "one search region with four templates" token organization. For each modality $m \in \{\mathrm{rgb}, \mathrm{tir}\}$, the search region and the four templates are individually tokenized with positional encodings and then concatenated into a unified token sequence

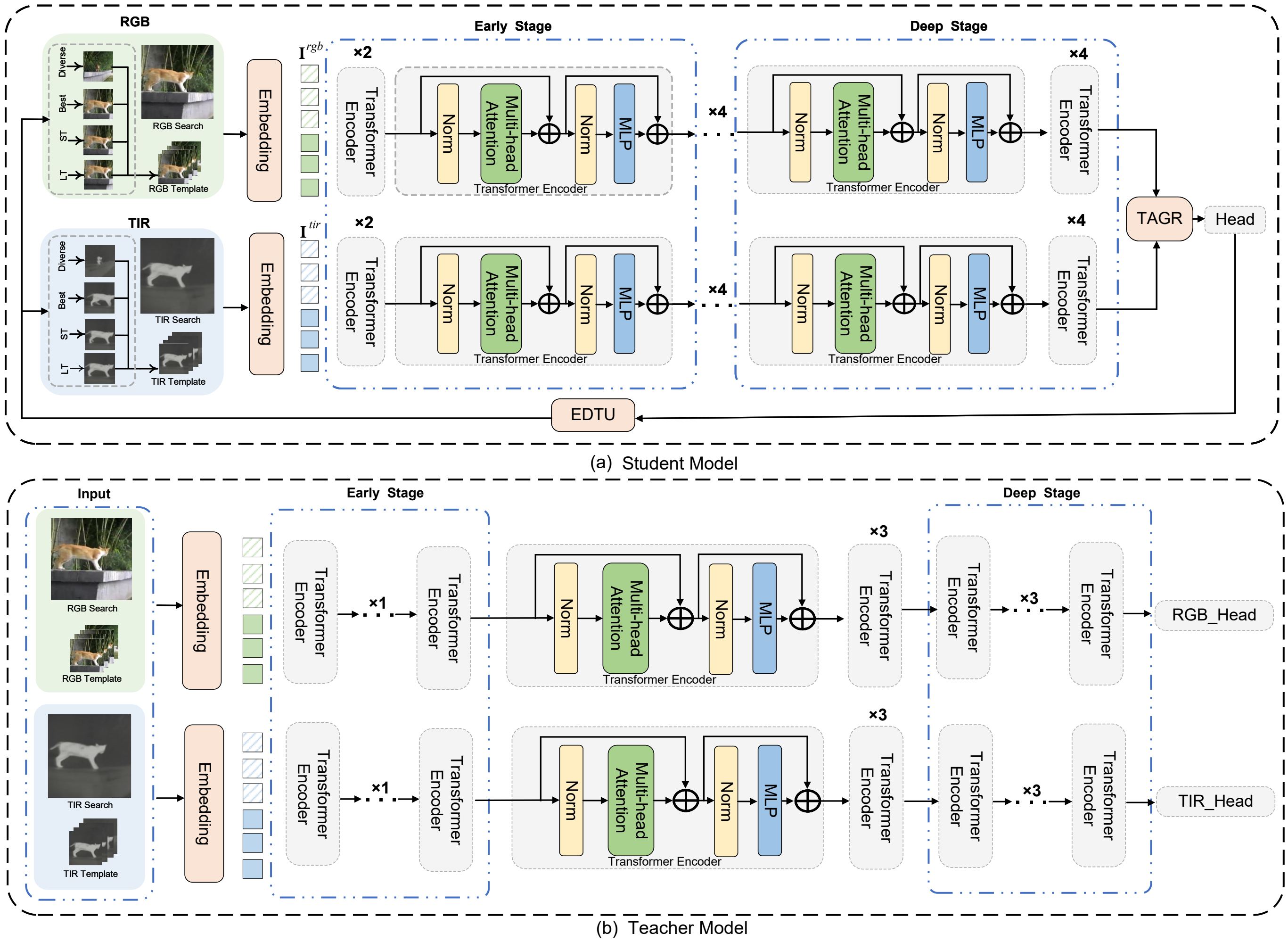

Fig. 4. hlOverall framework of D³-Track. (a) Student model: RGB/TIR search–template tokens are processed by modality-specific ViT encoders (Early/Deep stages), fused by TAGR, and decoded by a multimodal head; EDTU performs event-driven template updates at inference. (b) Teacher model: Two unimodal teachers (RGB_Head/TIR_Head) provide bidirectional cross-supervision to the students via HCKD at selected Early/Deep layers. Teachers and unimodal heads are used only during training; inference keeps (a) only.

$\mathbf{I}_t^m = [S_t^m; T_{t,1}^m; T_{t,2}^m; T_{t,3}^m; T_{t,4}^m]$. During feature extraction, each modality-specific token sequence is fed into its corresponding ViT backbone. At the training stage, we first perform Hierarchical Cross-modal Knowledge-alignment Distillation (HCKD) across multiple levels to enhance the students' representational capacity. Then, the TAGR module is adopted to adaptively fuse the bimodal features. The fused feature representation is finally propagated to the multimodal tracking head to produce classification and regression results.

B. Hierarchical Cross-modal Knowledge-Alignment Distillation

To enable effective knowledge transfer between RGB and TIR modalities, we propose the Hierarchical Cross-Modal Knowledge Alignment Distillation (HCKD). The overall pipeline is illustrated in Fig. 5. It adopts a cross-supervision scheme where each modality-specific student is guided by the teacher from the other modality, achieving bidirectional knowledge transfer while aligning representations across different layers. Although the teacher and student are optimized jointly with the same ViT-B backbone, the teacher is trained only with its unimodal tracking head , while the student is optimized with a multimodal tracking head and additionally receives the teacher's cross-modal distillation/alignment guidance; the teacher is used only during training and discarded at inference, introducing no extra complexity.

HCKD follows a progressive strategy of "align first, then transfer". Its core idea is to build a consistent cross-modal representation space via layer-wise feature alignment, upon which high-level discriminative ability can be effectively transferred. Following this idea, the specifics of each stage are elaborated below.

Layer-wise alignment. To construct a consistent space, we align features at multiple levels. Given teacher features $\mathrm{F}_t \in \mathbb{R}^{B\times N\times C}$ and student features $\mathrm{F}_s \in \mathbb{R}^{B\times N\times C}$ (batch B, tokens N, channels C), alignment is performed independently at different layers.

Early (statistical) alignment. We match channel-wise statistics with a moment-matching loss:

$$\mathcal{L}_{\text{stat}} = \frac{1}{C}\sum_{c=1}^{C}\Big[(\mu_c^t - \mu_c^s)^2 + (\sigma_c^t - \sigma_c^s)^2\Big], \tag{1}$$

where μ_c^t and μ_c^s denote the mean of the c-th channel for teacher and student, and σ_c^t and σ_c^s are the corresponding stan-

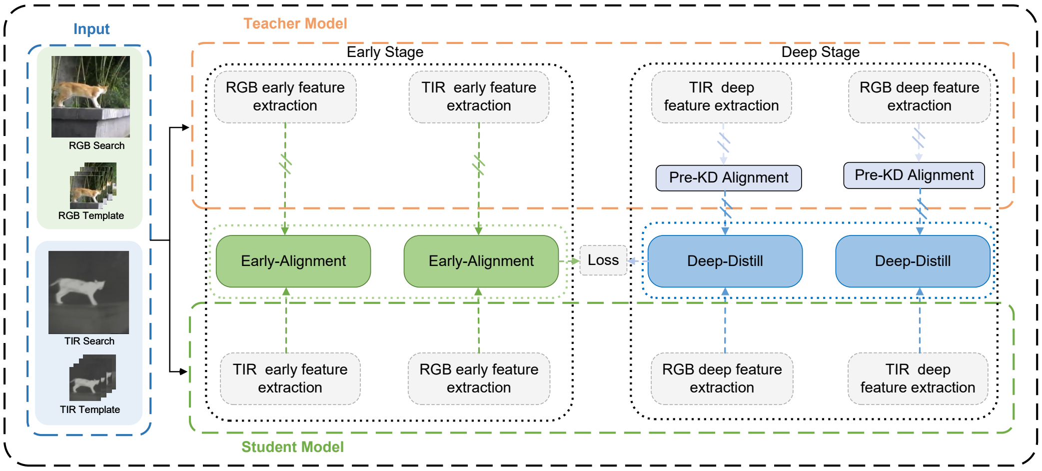

Fig. 5. HCKD: Cross-supervision framework for hierarchical cross-modal knowledge alignment. In the Early Stage, RGB and TIR teachers supervise the opposite students via Early-Alignment on shallow features. In the Deep Stage, Pre-KD Alignment projects teacher/student features to a shared space and Deep-Distill transfers high-level semantics. Losses are applied at selected layers; teachers and unimodal heads are used only during training.

dard deviations. It provides a stable shallow-layer alignment by reducing modality-specific distribution bias (mean/variance) before semantic transfer.

Deep (projection & semantic) distillation. We first unify the feature subspace using a learnable projection head, referred to as Pre-KD Alignment:

$$F_t^{\text{proj}} = W\, F_t + b, \tag{2}$$

This learnable mapping alleviates cross-modal subspace mismatch and avoid directly applying feature regression losses for hard alignment on raw features, which would force the student to mimic the teacher point-by-point and suppress its own modality-specific characteristics. where $W \in \mathbb{R}^{C_s \times C_t}$ and $b \in \mathbb{R}^{C_s}$ map teacher channels to the student space. After subspace alignment, we apply instance-wise normalization to remove scale differences:

$$F_t^{\text{norm}} = \frac{F_t^{\text{proj}} - \mu(F_t^{\text{proj}})}{\sigma(F_t^{\text{proj}})}, \qquad F_s^{\text{norm}} = \frac{F_s - \mu(F_s)}{\sigma(F_s)}, \tag{3}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ are computed along the token dimension. We then impose directional consistency in the shared semantic space:

$$\mathcal{L}_{\text{dir}} = 1 - \frac{1}{BN} \sum_{i=1}^{B} \sum_{j=1}^{N} \frac{\left\langle F_{t,ij}^{\text{norm}},\, F_{s,ij}^{\text{norm}} \right\rangle}{\left\| F_{t,ij}^{\text{norm}} \right\|_2 \left\| F_{s,ij}^{\text{norm}} \right\|_2}. \tag{4}$$

By matching feature directions (cosine similarity) rather than magnitudes, $\mathcal{L}_{\text{dir}}$ transfers high-level discriminative semantics while being less sensitive to cross-modal scale variations, thereby preserving the student's modality-specific structure.

The HCKD loss is the weighted sum:

$$\mathcal{L}_{\text{HCKD}} = \lambda_{\text{stat}} \mathcal{L}_{\text{stat}} + \lambda_{\text{dir}} \mathcal{L}_{\text{dir}}, \tag{5}$$

where λ_{stat} and λ_{dir} are balancing hyperparameters. During training, we cache intermediate features from each Transformer encoder block to support end-to-end layer-wise optimization. With this progressive alignment and transfer, HCKD improves the student's representation quality and generalization ability.

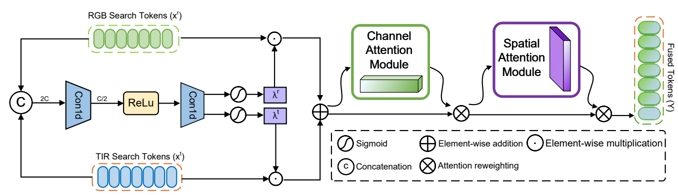

Fig. 6. Illustration of TAGR. RGB and TIR search tokens are fed into a lightweight conv–ReLU–conv branch to estimate per-token modality weights (λ^r, λ^t) under the complementary constraint $\lambda^r + \lambda^t = 1$. Using these weights, TAGR performs token-wise convex fusion of the two modalities, followed by CA and SA for post-fusion feature refinement.

C. Token-wise Adaptive Gating & Recalibration

In RGB-T tracking, the reliability of RGB and TIR cues can vary across space and time. To directly adapt to such variations, our Token-wise Adaptive Gating & Recalibration (TAGR) module takes the current-frame RGB and TIR search tokens, concatenates them into joint evidence, and predicts token-wise complementary mixing weights (λ^r, λ^t). These weights are then used to perform convex fusion of the two modalities before channel–spatial recalibration, as illustrated in Fig. 6.

The detailed implementation of TAGR is described as follows:

At the input of the fusion block, the RGB search feature $x^r \in \mathbb{R}^{B \times N \times C}$ and the TIR search feature $x^t \in \mathbb{R}^{B \times N \times C}$ are concatenated along the channel dimension to form

$$U = [x^r;\, x^t] \in \mathbb{R}^{B \times N \times 2C}.$$

We apply two 1D convolutions along the token dimension (with channel transitions $2C \to C/2 \to 1$) to generate the gating logits $g \in \mathbb{R}^{B \times 1 \times N}$. Applying a Sigmoid yields the token-wise reliability of the RGB modality,

$$\lambda^r = \sigma(g) \in \mathbb{R}^{B \times 1 \times N}, \qquad \lambda^t = 1 - \lambda^r.$$

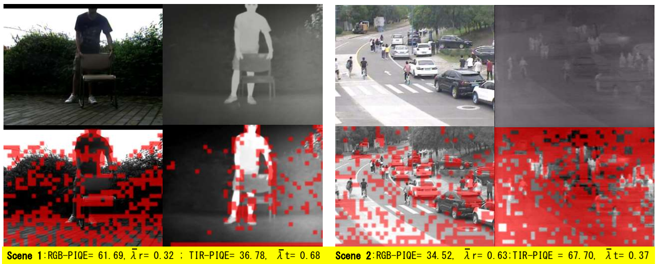

Fig. 7. PIQE-based block-level quality visualization (offline analysis only). Top: original frames. Bottom: red overlays indicate blocks with higher PIQE scores (darker = more degradation). Left to right: RGB and TIR for Scene 1 and Scene 2.

The two modalities are then fused by weighted aggregation:

$$F = \lambda^r \odot x^r + \lambda^t \odot x^t.$$

To strengthen the discrimination of the fused representation, we further apply an attention-based channel–spatial recalibration to the fused tokens $F \in \mathbb{R}^{B\times N\times C}$. First, global average pooling along the token dimension followed by a bottleneck 1×1 conv ($C \to \frac{C}{2} \to C$) produces a channel attention map $A_{\mathrm{c}} \in \mathbb{R}^{B\times 1\times C}$, which is used to rescale each channel of F:

$$F_{\mathrm{c}} = A_{\mathrm{c}} \odot F,$$

where F_{c} denotes the channel-recalibrated fused representation and $\odot$ is element-wise multiplication with broadcasting along the token dimension. Second, the average pooling and max pooling operations along the channel dimension are applied to F_c to obtain two channel descriptors. Then, the two descriptors are concatenated and fed to a 1D convolution with kernel size 3 to obtain a spatial (token-wise) attention map $A_{\mathrm{s}} \in \mathbb{R}^{B\times N\times 1}$, finally yielding the token-enhanced output Y:

$$Y = A_{\mathrm{s}} \odot F_{\mathrm{c}}.$$

The final fused tokens Y are sent to the tracking head for subsequent target localization.

In Fig. 7, we visualize the Perception based Image Quality Evaluator (PIQE) based quality of RGB and TIR images. PIQE is only used for visualization, not inference. Higher PIQE indicates worse perceptual quality and more noise. TAGR's adaptive weighting strategy consistently emphasizes the modality with better local quality—assigning higher weights to TIR ($\bar{\lambda}_t$ = 0.68)) where it outperforms RGB (Scene 1), and favoring RGB ($\bar{\lambda}_r$ = 0.63)) where it is cleaner (Scene 2). This demonstrates TAGR's capability in effectively suppressing modality-quality noise.

D. Event-Driven Template Update

To mitigate temporal bias arising from template anchoring and unstable updates, we propose an Event-Driven Template Update (EDTU) mechanism (see Algorithm 1 in appendix). EDTU maintains four functionally distinct templates, forming a library $\mathcal{T} = \{T_{\mathrm{LT}}, T_{\mathrm{ST}}, T_{\mathrm{Best}}, T_{\mathrm{Div}}\}$, and replaces fixed-interval updates with event-triggered operations for more robust template management. To visualize the effect of EDTU, Fig. 8(a) plots the decay in similarity to the first-frame template as the frame distance increases, revealing a clear

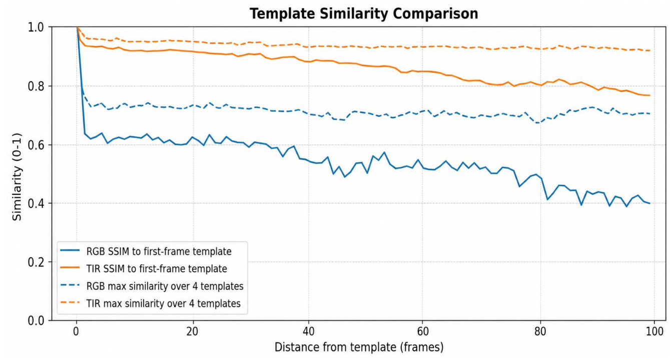

Fig. 8. Visualization of temporal bias mitigation using the multi-template pool. SSIM is used to measure appearance consistency. Solid curves show the similarity to the first-frame template, which decreases with frame distance and reflects temporal bias. Dashed curves show the maximum similarity to the four templates maintained by EDTU (LT/ST/Best/Diverse), which remains more stable over time. This demonstrates that the multi-template pool alleviates template staleness and maintains more reliable target representations.

temporal drift. In contrast, Fig. 8(b) shows that this drift is effectively mitigated when selecting the maximum similarity among the four template slots at each frame, resulting in a much flatter curve. This comparison demonstrates EDTU's capability in maintaining template relevance over time.

This capability is grounded in a structured template library, which is initialized at the first frame. The roles of the four template slots are defined as follows:

- **Long-term (LT):** a stable anchor fixed to the initial template; never updated; used for fallback and drift suppression.
- **Short-term (ST):** rapidly adapts to recent appearance changes, remaining sensitive to dynamic scenes.
- **Best:** stores the highest-confidence template and serves as the reference for quality and difference measures.
- **Diversity (Div):** captures complementary patterns under pose/scale changes or partial occlusions to enhance diversity.

Given the response map $R_t \in \mathbb{R}^N$ at frame t, we extract three lightweight scalar cues to characterize its instantaneous characteristics: a *quality* score Q_t, a *stability* score S_t, and a *novelty* score N_t.

Quality. The quality score Q_t reflects both the peak strength and the concentration of the response. Specifically, we define the peak magnitude $c_t = \max(R_t)$ and a concentration term

$$p_t = \frac{\mathrm{Top1}(R_t)}{\mathrm{mean}(\mathrm{Top}\,k(R_t))}, \quad k = 5,$$

where $\mathrm{Top}\,k(\cdot)$ denotes the k highest responses. Their product gives the final quality measure:

$$Q_t = c_t \cdot p_t \tag{6}$$

Stability. To measure short-term temporal consistency, we compute the IoU between the current predicted box b_t and the previous one b_{t-1}:

$$S_t = \mathrm{IoU}(b_t, b_{t-1}). \tag{7}$$

A low S_t indicates that the tracker is unstable or drifting.

Novelty. We compute the negative cosine similarity between the current response and two references—the previous-frame response and the best-template response—and take the larger value as the final novelty N_t:

$$\begin{aligned} N_t^{\text{short}} &= 1 - \cos\big(z(R_t),\, z(R_{t-1})\big), \\ N_t^{\text{best}} &= 1 - \cos\big(z(R_t),\, z(R_t^{\text{Best}})\big), \\ N_t &= \max\big(N_t^{\text{short}},\, N_t^{\text{best}}\big). \end{aligned} \tag{8}$$

Here, R_t^{best} is the response map obtained at frame t using the best historical template. The operator $z(\cdot)$ applies per-map z-score normalization, and $\cos(\cdot,\cdot)$ denotes cosine similarity. Accordingly, N_t^{short} and N_t^{best} measure the cosine distances to the previous frame and the best template, respectively. A larger N_t indicates a more novel pattern compared with recent history or the best template.

Event score and stability gating. The three cues are combined into a single event score while enforcing a stability gate:

$$\Phi_t = Q_t\, N_t\, g(S_t), \qquad g(S_t) = \begin{cases} 0, & S_t < 0.5, \\ 1, & \text{otherwise}, \end{cases} \tag{9}$$

Consequently, template updates are disabled when the tracker is clearly unstable.

Adaptive threshold. To avoid a fixed, hand-crafted threshold, we use a sliding quantile over the latest M event scores as an adaptive threshold:

$$\tau_t = \text{Quantile}_q\left(\{\Phi_{t-j}\}_{j=1}^{M}\right), \tag{10}$$

with M=30 and q=0.7 in our experiments. Whenever $\Phi_t > \tau_t$, the current template is pushed into the short-term (ST) slot.

Best template with temporal aging. To alleviate the "locking" effect of early high-quality samples, we apply a temporal aging strategy to the stored best score:

$$\hat{Q}^{\text{Best}} \leftarrow (1-d)\,\hat{Q}^{\text{Best}}, \qquad \text{update Best if} \;\; Q_t > \hat{Q}^{\text{Best}}, \tag{11}$$

where d is the aging rate. Only when the current quality score surpasses the decayed best score do we refresh the Best template.

Diversity-triggered slot. The update of the Div template is driven by appearance difference: when the cosine similarity between the current frame and the Best template is below the threshold δ_{div}, or the temporal stability $S_t < \delta_{\text{stab}}$, and the current quality $Q_t \geq \kappa Q^{\text{Best}}$, an update operation is triggered to expand the diversity of the template library. Fig. 9 provides an example with non-rigid cat motion, where the target undergoes noticeable deformation and appearance change. The Div slot is updated at frame t (marked by $\star$), storing a complementary template that captures the deformed appearance, which helps maintain stable tracking in subsequent frames.

E. Final Loss

The overall training objective combines the standard tracking task loss and the proposed HCKD loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}}^{mm} + \mathcal{L}_{\text{task}}^{t,rgb} + \mathcal{L}_{\text{task}}^{t,tir} + \lambda_{\text{HCKD}}\, \mathcal{L}_{\text{HCKD}}, \tag{12}$$

where $\mathcal{L}_{\text{task}}^{mm}$ denotes the multimodal tracking loss of the student model, and $\mathcal{L}_{\text{task}}^{t,rgb}$ and $\mathcal{L}_{\text{task}}^{t,tir}$ denote the unimodal tracking losses for the RGB and TIR teacher branches, respectively. Following OSTrack [21].

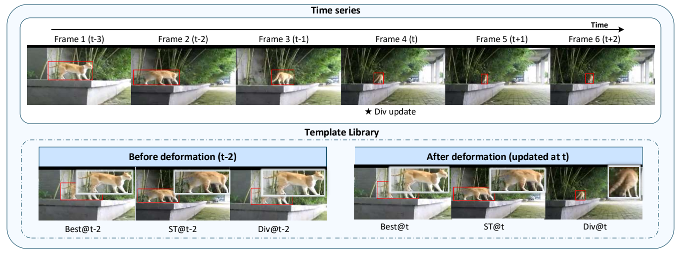

Fig. 9. Qualitative analysis of the diversity-triggered template slot (Div) in EDTU under non-rigid deformation. Top: tracking results on a *cat* sequence before and after deformation with template update; the Div update is triggered at frame t (marked by $\star$). Bottom: snapshots of the template library before deformation ($t-2$) and after deformation (updated at t), where Div stores a complementary template that captures the deformed appearance. This diverse template helps maintain tracking stability when the Best template becomes outdated and the Short-Term template may be unreliable.

$\mathcal{L}_{\text{HCKD}}$ is defined in Eq. (5), and is applied hierarchically across multiple Transformer blocks. The teacher branches are optimized with unimodal tracking losses, and the distillation gradients are stopped at the teacher side (i.e., no back-propagation into the teachers through HCKD).

IV. Experiment

A. Implementation Details

Our implementation is based on OSTrack [21] with a ViT-B backbone. The backbone was initialized with pre-trained weight from DropMAEs [22]. The entire model was trained end-to-end with AdamW, minimizing a joint objective of classification and regression losses. During training, we adopt a "one search with four templates" token organization, where three supplementary templates are randomly sampled in addition to the primary template–search pair to improve robustness under heterogeneous template quality. During inference, we employ the Event-Driven Template Update (EDTU) mechanism to maintain a template pool online based on quality, stability, and novelty cues.We trained our model on the LasHeR training set and directly evaluated it on the full RGBT234 test set without any data-specific fine-tuning. For the VTUAV dataset, we trained on its complete training split and validated on the short-term test subset. The learning rates for the backbone and the tracking head were set to 5×10^{-6} and 5×10^{-5}, respectively. All models were implemented in PyTorch and trained on two NVIDIA RTX 4090D GPUs (24GB) with a global batch size of 6. Fine-tuning was conducted for 30 epochs, each comprising 60,000 training pairs.

B. Datasets and Evaluation Metrics

We perform a comprehensive evaluation on three RGB-T benchmarks (RGBT234 [23], LasHeR [24], and VTUAV [25]) using standard metrics:

Precision: the percentage of frames whose predicted box center lies within a fixed threshold from the ground truth.

Normalized Precision [26], [27]: a scale-invariant metric that normalizes the center error by the target size.

TABLE I
OVERALL COMPARISON ON LASHER, RGBT234, AND VTUAV-ST. "—" MEANS NOT REPORTED.

Method	Source	LasHeR			RGBT234		VTUAV-ST	
		PR	NPR	SR	MPR	MSR	MPR	MSR
D^3-Track-B384	Ours	80.2	77.2	62.9	93.2	68.0	87.8	75.7
D^3-Track-B256	Ours	78.9	76.2	61.8	93.6	69.7	91.9	78.3
CADTrack	AAAI 2026 [28]	77.7	73.3	61.3	92.8	67.7	—	—
GOLA	AAAI 2026 [29]	77.5	73.9	61.6	92.2	69.5	—	—
MFDSR	TCSVT 2026 [30]	47.5	—	39.5	91	68.2	—	—
RAGTrack	CVPR 2026 [31]	76.8	73	61.1	93.8	69.5	—	—
ProMoT	TCSVT 2026 [32]	76.3	72.6	60.9	83.7	58.2	77.3	62.2
VCT	TMM 2026 [33]	74.8	—	59.3	91.3	67.9	—	—
PromptTrack	TMM 2026 [34]	76.2	—	61.4	91.7	67.2	—	—
MPANet	TMM 2025 [35]	72.1	—	57.3	87.4	64.6	—	—
FMTrack	TCSVT 2025 [36]	75.6	71.7	59.7	85.9	64.9	85.7	72.8
TUMFNet	IJCAI 2025 [37]	76.4	72.7	61.4	90.8	67.8	—	—
STTrack	AAAI 2025 [38]	76.0	—	60.3	89.8	66.7	—	—
AETrack	TCSVT 2025 [39]	74.7	71.0	59.6	91.6	68.8	—	—
$TBSI_{ext}$	TPAMI 2025 [14]	75.5	71.5	59.6	91.0	67.0	—	—
AINet	AAAI 2025 [13]	74.2	70.1	59.1	89.2	67.3	84.7	72.9
DMD	TIP 2025 [40]	72.6	68.6	57.6	89.3	66.7	—	—
TVTracker	IoT-J 2025 [41]	72.6	68.4	57.5	88.6	64.8	—	—
CKD	ACMMM 2024 [16]	73.2	69.3	58.1	90.0	67.4	—	—
GMMT	AAAI 2024 [42]	70.7	67.0	56.6	87.9	64.7	—	—
BAT	AAAI 2024 [43]	70.2	—	56.3	86.8	64.1	82.6	68.5
TATrack	AAAI 2024 [44]	70.2	66.7	56.1	87.2	64.4	—	—
CAFormer	AAAI 2025 [9]	70.0	66.1	55.6	88.3	66.4	—	—
TBSI	CVPR 2023 [7]	69.2	65.7	55.6	87.1	63.7	84.4	72.6
CMDTrack	CVPR 2023 [45]	68.8	—	56.6	85.9	61.8	—	—
Un-Track	CVPR 2024 [46]	66.7	—	53.6	83.7	61.8	—	—
SDSTrack	CVPR 2024 [47]	66.5	—	53.1	84.8	62.5	—	—
ViPT	CVPR 2023 [48]	65.1	—	52.5	83.5	61.7	86.0	74.1

Success Rate: the percentage of frames whose IoU exceeds a given threshold, where the area under the success curve is reported as the overall success score.

To further reduce the bias introduced by slight annotation misalignment, following HMFT [25], we additionally report on VTUAV the Max Precision Rate (MPR) and Max Success Rate (MSR) as complementary indicators.

C. Comparison with Competing Methods

We comprehensively evaluate D^3-Track on three benchmark RGB-T datasets: LasHeR [24], RGBT234 [23], and VTUAV [25], and compare it against 26 state-of-the-art trackers. Quantitative results are summarized in Table I.

On RGBT234.RGBT234 is a standard RGB-T tracking benchmark with 234 pairs of aligned RGB-T sequences, containing about 233.4K frames. As reported in Table I, D^3-Track shows strong competitiveness on this benchmark. D^3-Track-B256 achieves 93.6 MPR and 69.7 MSR, delivering the best success performance among all compared trackers and a comparable precision score to the best-performing method. In particular, the higher MSR indicates that D^3-Track produces more accurate bounding boxes under different overlap thresholds, demonstrating the effectiveness of the proposed denoising-driven design for robust RGB-T tracking.

On LasHeR. We further evaluate the robustness of D^3-Track on the large-scale LasHeR benchmark. LasHeR is currently one of the largest and most challenging RGB-T tracking datasets, comprising 1,224 pairs of RGB-T sequences with approximately 734.8K frames. It also provides annotations for 19 challenging attributes, such as HI (high illumination) and AIV (abrupt illumination variation), making it suitable for evaluating tracker robustness under complex scenarios.

As shown in Table I, we compare D^3-Track with 26 state-of-the-art RGB-T trackers using PR, NPR, and SR. D^3-Track-B384 achieves the best results on all three metrics, reaching 80.2/77.2/62.9 in PR/NPR/SR. Compared with the best existing trackers under each metric, D^3-Track-B384 improves PR, NPR, and SR by 2.5, 3.3, and 1.3 percentage points, respectively. Meanwhile, D^3-Track-B256 also achieves highly competitive performance with 78.9/76.2/61.8, surpassing all existing trackers reported in the table. These results demonstrate that the proposed denoising-driven design effectively improves tracking robustness on the challenging LasHeR benchmark.

On VTUAV. The VTUAV dataset collects RGB-T data from UAV scenarios and broadens the application scope of RGB-T tracking. It contains 500 aligned RGB-T video sequences with up to 1.7 million frames. Our experiments focus on its short-term tracking subset. As shown in Table I, D^3-Track-B256 achieves the best performance on VTUAV-ST, obtaining 91.9 MPR and 78.3 MSR. Compared with the best existing tracker under each metric, D^3-Track-B256 improves MPR and MSR by 5.9 and 4.2 percentage points, respectively. These results demonstrate the strong robustness and generalization capability of D^3-Track in UAV-based RGB-T tracking scenarios, where targets often undergo scale variation, viewpoint changes, and background interference.

We also observe that D^3-Track-B384 performs better on LasHeR, while D^3-Track-B256 achieves higher results on RGBT234 and VTUAV-ST. This suggests that increasing the input resolution does not always bring consistent gains. Compared with LasHeR, RGBT234 and VTUAV-ST contain more cases where additional high-resolution tokens may introduce redundant background context or irrelevant cross-modal responses. Therefore, the 256×256 input provides a more compact representation and achieves a better balance between target discrimination and background suppression on these two benchmarks.

TABLE II
EFFICIENCY COMPARISON WITH REPRESENTATIVE RGB-T TRACKERS ON LASHER. FPS IS MEASURED ON AN RTX 4090D GPU.

Method	PR↑	NPR↑	SR↑	Params↓	FLOPs↓	FPS↑
D^3-Track (Ours)	**78.9**	**76.2**	**61.8**	185.4M	91.1G	**116.51**
CADTrack [28]	77.7	73.3	61.3	130.1M	77.4G	87.7
TBSI [7]	69.2	65.7	55.6	201.9M	82.5G	108.54
FMTrack [36]	75.6	71.7	59.7	286.4M	128.3G	56.4
RAGTrack [31]	76.8	73.0	61.1	**111.4M**	**62.7G**	54.4

D. *Efficiency Analysis*

Table II compares the accuracy-efficiency trade-off of representative RGB-T trackers on LasHeR. D3-Track achieves the highest PR/NPR/SR scores of 78.9/76.2/61.8 and runs at 116.51 FPS on an RTX 4090 GPU. Compared with CADTrack, D3-Track improves PR/NPR/SR by +1.2/+2.9/+0.5 points while being faster in inference. Although RAGTrack has the lowest parameter count and FLOPs, its FPS is considerably lower than that of D3-Track, suggesting that end-to-end tracking speed is also affected by the overall inference pipeline rather than only by theoretical FLOPs. Benefiting from the removal of auxiliary teacher branches during inference and the lightweight token-wise fusion design, D3-Track achieves a favorable balance between tracking accuracy and real-time efficiency.

E. Ablation Studies

The template update mechanism serves as a critical bottleneck in multimodal tracking, as it dictates the reliability of the memory bank used for subsequent fusion and distillation. In challenging scenarios, a single erroneous update can propagate corrupted templates through the entire pipeline, fundamentally compromising the evaluation of other components. Therefore, on the LasHeR dataset, we first conduct an ablation of update strategies to establish the effectiveness of the proposed EDTU mechanism (see Table III). On this basis, we further evaluate the contributions of the TAGR and HCKD modules, thereby avoiding systemic bias introduced by template distortion and ensuring that the performance gains of each component are attributable and reproducible.

1) Module-wise Ablation: **Baseline.** We remove HCKD, TAGR, and EDTU from the full pipeline while retaining the four-template input form. During inference, template updating is disabled, and the first-frame template is repeatedly used to fill the four template slots. This setting serves as the baseline for evaluating different online template update strategies.

Periodic Update. We first evaluate a simple periodic updating strategy, where the dynamic template is refreshed every fixed number of frames without considering the reliability of the current prediction. As shown in Table III, Periodic Update obtains PR/NPR/SR of 72.4/69.5/57.0. Although it slightly improves PR by +0.5 points over the Baseline, it decreases NPR by 0.5 points and brings no improvement in SR. This indicates that fixed-interval updating may introduce unreliable templates when the current prediction is inaccurate.

SGTU (Score-Gated Template Update). We further compare EDTU with a confidence-based updating strategy, following the score-gated template update policy used in STARK [49] and MixFormer [50]. We refer to this variant as Score-Gated Template Update (SGTU). The first-frame template is kept as a fixed anchor, and a first-in, first-out (FIFO) queue of dynamic templates is maintained. A new template is enqueued only when the predicted confidence exceeds a preset threshold. Apart from the updating policy, all other settings are kept the same for fair comparison. SGTU achieves PR/NPR/SR of 74.7/72.3/59.1, yielding absolute gains of +2.8/+2.3/+2.1 points over the Baseline. This confirms the benefit of confidence-gated dynamic updating, but it still relies on a single confidence cue and lacks explicit consideration of appearance novelty and temporal stability.

w/ EDTU. Replacing SGTU with our EDTU further improves PR/NPR/SR to 76.7/74.2/60.2, corresponding to +4.8/+4.2/+3.2 points over the Baseline and +2.0/+1.9/+1.1 points over SGTU. Compared with Periodic Update and SGTU, EDTU jointly considers quality, novelty, and temporal stability cues, and manages templates with ST, Best, and Div slots. These results demonstrate that event-driven, evidence-aggregated template updating is more effective than fixed-interval or confidence-thresholding strategies in RGB-T tracking.

TABLE III
EFFECTIVENESS OF THE EDTU MECHANISM ON **LASHER**. BEST RESULTS IN BOLD AND SECOND-BEST UNDERLINED.

Method	PR	NPR	SR	Avg. Δ
Baseline	71.9	70.0	57.0	–
Periodic Update	$72.4_{\uparrow 0.5}$	$69.5_{\downarrow 0.5}$	$57.0_{\uparrow 0.0}$	+0.0
SGTU	$\underline{74.7}_{\uparrow 2.8}$	$\underline{72.3}_{\uparrow 2.3}$	$\underline{59.1}_{\uparrow 2.1}$	<u>+2.4</u>
w/ EDTU	$\mathbf{76.7}_{\uparrow 4.8}$	$\mathbf{74.2}_{\uparrow 4.2}$	$\mathbf{60.2}_{\uparrow 3.2}$	**+4.1**

TABLE IV
COMPONENT ANALYSIS OF TAGR ON LASHER. BEST RESULTS IN BOLD AND SECOND-BEST UNDERLINED.

Method	PR	NPR	SR	Avg. Δ
w/o TAGR	77.3	75.2	<u>61.3</u>	–
w/o Token-wise Gating	$76.8_{\downarrow 0.5}$	$73.8_{\downarrow 1.4}$	$60.0_{\downarrow 1.3}$	-1.1
w/o Post-fusion Recalibration	$\underline{77.7}_{\uparrow 0.4}$	$\underline{75.4}_{\uparrow 0.2}$	$61.0_{\downarrow 0.3}$	<u>+0.1</u>
Full TAGR	$\mathbf{78.9}_{\uparrow 1.6}$	$\mathbf{76.2}_{\uparrow 1.0}$	$\mathbf{61.8}_{\uparrow 0.5}$	**+1.0**

Module-wise Ablation (EDTU fixed). With EDTU enabled, we progressively introduced TAGR and HCKD to quantify their individual contributions, as shown in Table VI.

w/ EDTU+TAGR. Building upon the EDTU baseline, adding the TAGR module improves PR/NPR/SR by +1.0%/+1.0%/1.1%, confirming the effectiveness of TAGR.

w/ EDTU+HCKD. Introducing the HCKD module accompanying EDTU yields additional gains of +0.6%/+0.4%/+0.6% on PR/NPR/SR.

w/ EDTU+TAGR+HCKD (Full). The full model integrating EDTU, TAGR, and HCKD achieves the best performance, delivering +2.2%/+2.0%/+1.6% improvements over the EDTU-only configuration on PR/NPR/SR, indicating that TAGR and HCKD are complementary and produce a synergistic enhancement when combined with EDTU.

2) Comparison with Representative Fusion Modules: To further verify the effectiveness of TAGR, we compare it with representative fusion designs under the same framework, while keeping HCKD and EDTU unchanged. Specifically, CBAM-based Fusion first combines RGB and TIR features and then applies CBAM for attention-based recalibration, while AFF [51] performs attentional feature fusion through learned fusion weights. As shown in Table V, TAGR achieves the best PR/NPR/SR scores of 78.9/76.2/61.8, outperforming CBAM-based Fusion and AFF. This indicates that generic attention recalibration or general-purpose attentional feature fusion is less effective for RGB-T tracking, where modality reliability varies across spatial tokens. In contrast, TAGR explicitly estimates token-wise complementary weights and further refines the fused representation, leading to more reliable modality integration.

3) TAGR Component Analysis: As shown in Table IV, removing either token-wise gating or post-fusion recalibration degrades the performance, confirming the contribution of both operations. Notably, the variant without token-wise gating performs even worse than directly removing TAGR. This is because post-fusion recalibration relies on a reliability-aware fused representation. Without adaptive modality weighting, low-quality modality cues may be directly mixed into the fused

TABLE V
COMPARISON WITH REPRESENTATIVE FUSION MODULES ON LASHER. BEST RESULTS IN BOLD AND SECOND-BEST UNDERLINED.

Method	PR	NPR	SR	Avg. Δ
CBAM-based Fusion	76.4	73.5	59.6	–
AFF [51]	$\underline{78.0}_{\uparrow 1.6}$	$\underline{74.0}_{\uparrow 0.5}$	$\underline{60.3}_{\uparrow 0.7}$	$\underline{+0.9}$
TAGR	$\mathbf{78.9}_{\uparrow 2.5}$	$\mathbf{76.2}_{\uparrow 2.7}$	$\mathbf{61.8}_{\uparrow 2.2}$	**+2.5**

TABLE VI
MODULE-WISE ABLATION ON LASHER. BEST RESULTS IN BOLD AND SECOND-BEST UNDERLINED.

Method	PR	NPR	SR	Avg. Δ
w/ EDTU	76.7	74.2	60.2	–
w/ EDTU + TAGR	$\underline{77.7}_{\uparrow 1.0}$	$\underline{75.2}_{\uparrow 1.0}$	$\underline{61.3}_{\uparrow 1.1}$	$\underline{+1.0}$
w/ EDTU + HCKD	$77.3_{\uparrow 0.6}$	$74.6_{\uparrow 0.4}$	$60.8_{\uparrow 0.6}$	+0.5
w/ EDTU + TAGR + HCKD	$\mathbf{78.9}_{\uparrow 2.2}$	$\mathbf{76.2}_{\uparrow 2.0}$	$\mathbf{61.8}_{\uparrow 1.6}$	**+1.9**

tokens, and the subsequent recalibration may further emphasize these unreliable responses. This indicates that token-wise gating is a key prerequisite for effective recalibration, while the full TAGR achieves the best performance by combining adaptive modality weighting with post-fusion feature refinement.

4) HCKD Component Analysis: To further validate the effectiveness of HCKD, we conduct a fine-grained ablation study on Early Alignment, Pre-KD Alignment, and Deep Distillation, as shown in Table VII. The variant with only Early Alignment is used as the reference for computing the gains.

Only Early Alignment. With only Early Alignment, the model achieves PR/NPR/SR of 71.6/69.5/56.9. This indicates that shallow statistical alignment provides a basic cross-modal normalization effect, but it is insufficient to transfer deeper discriminative knowledge.

w/o Pre-KD Alignment. When Pre-KD Alignment is removed, Deep Distillation is directly applied on top of Early Alignment. The performance improves to 72.5/70.2/57.5, with an average gain of +0.7 points over the reference. This shows that deep distillation can provide additional high-level supervision, but direct distillation without a proper aligned feature space still limits its effectiveness.

w/o Early Alignment. When Early Alignment is removed while Pre-KD Alignment and Deep Distillation are retained, the model obtains 72.3/70.2/57.3, with an average gain of +0.6 points. The result suggests that Pre-KD Alignment and Deep Distillation are useful for cross-modal knowledge transfer, while shallow statistical alignment further helps stabilize the subsequent transfer process.

Full HCKD. With all three components enabled, HCKD achieves the best performance of 73.3/70.9/58.3, improving the reference by +1.7/+1.4/+1.4 points. Compared with the variants that remove either Early Alignment or Pre-KD Alignment, the full model consistently obtains higher scores, demonstrating that shallow statistical alignment, Pre-KD Alignment, and Deep Distillation are complementary. These results validate the proposed "align-then-transfer" design for improving RGB-T representation compatibility and cross-modal discriminative capability.

TABLE VII
FINE-GRAINED ABLATION OF HCKD ON LASHER. BEST RESULTS ARE SHOWN IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	PR	NPR	SR	Avg. Δ
Full HCKD	**73.3**	**70.9**	**58.3**	–
w/o Early Alignment	$72.3_{\downarrow 1.0}$	$\underline{70.2}_{\downarrow 0.7}$	$57.3_{\downarrow 1.0}$	-0.9
w/o Pre-KD Alignment	$\underline{72.5}_{\downarrow 0.8}$	$\underline{70.2}_{\downarrow 0.7}$	$\underline{57.5}_{\downarrow 0.8}$	$\underline{-0.8}$
Only Early Alignment	$71.6_{\downarrow 1.7}$	$69.5_{\downarrow 1.4}$	$56.9_{\downarrow 1.4}$	-1.5

TABLE VIII
TEMPLATE COMPONENT ABLATION OF EDTU ON LASHER. BEST IN BOLD; SECOND-BEST UNDERLINED.

ST	Best	Div	PR	NPR	SR	Avg. Δ
×	×	×	71.9	70.0	57.0	–
✓	×	×	$74.7_{\uparrow 2.8}$	$72.5_{\uparrow 2.5}$	$58.9_{\uparrow 1.9}$	+2.4
×	✓	×	$75.1_{\uparrow 3.2}$	$73.0_{\uparrow 3.0}$	$59.5_{\uparrow 2.5}$	+2.9
×	×	✓	$72.6_{\uparrow 0.7}$	$70.7_{\uparrow 0.7}$	$57.8_{\uparrow 0.8}$	+0.7
×	✓	✓	$76.1_{\uparrow 4.2}$	$73.8_{\uparrow 3.8}$	$\underline{59.9}_{\uparrow 2.9}$	+3.6
✓	×	✓	$76.1_{\uparrow 4.2}$	$73.8_{\uparrow 3.8}$	$59.6_{\uparrow 2.6}$	+3.5
✓	✓	×	$\underline{76.2}_{\uparrow 4.3}$	$\underline{74.1}_{\uparrow 4.1}$	$\underline{59.9}_{\uparrow 2.9}$	$\underline{+3.8}$
✓	✓	✓	$\mathbf{76.7}_{\uparrow 4.8}$	$\mathbf{74.2}_{\uparrow 4.2}$	$\mathbf{60.2}_{\uparrow 3.2}$	**+4.1**

5) EDTU Template Component Analysis: The EDTU mechanism handles target appearance variations by maintaining a dynamic template pool comprising a short-term (ST) template, a best (Best) template, and a diversity (Div) template. To dissect the specific contribution of each template component, we conducted detailed ablation experiments, with results summarized in Table VIII.

ST only (row 2). Compared with the no-update baseline (row 1), the ST-only strategy yields improvements of +2.8% (PR), +2.5% (NPR), and +1.9% (SR), indicating that short-term response is the basis of the gains.

Best only (row 3). It outperforms ST-only strategy. The results indicate that preserving high-quality, stable states across the tracking history yields a more reliable template (75.1%, 73.0%, 59.5%).

Div only (row 4). It brings limited gains (72.6%, 70.7%, 57.8%), suggesting that Div serves best as a complement to other methods, but is weak on its own.

Pairwise combinations (rows 5-7). Combining any two strategies gives clear benefits. Best+Div (76.1%, 73.8%, 59.9%) and ST+Div (76.1%, 73.8%, 59.6%) are strong. ST+Best achieves the best pairwise result (76.2%, 74.1%, 59.9%). This indicates that the combination of short-term response and the best historical template has the strongest synergy.

Full combination (row 8). Enabling all three strategies (ST+Best+Div) achieves the best performance (76.7%, 74.2%, 60.2%), demonstrating complementary and synergistic effects of our EDTU mechanism.

6) EDTU Cue Analysis: Table IX analyzes the contribution of different cues in EDTU. Removing any cue leads to performance degradation, indicating that quality, novelty, and stability are all useful for reliable template updating. Among them, removing the quality cue causes the most significant drop, suggesting that update quality is the primary factor for preventing unreliable templates from being introduced.

TABLE IX
CUE-LEVEL ABLATION OF EDTU ON LASHER. BEST RESULTS IN BOLD AND SECOND-BEST UNDERLINED.

Method	PR	NPR	SR	Avg. Δ
Full D^3-Track	**78.9**	**76.2**	**61.8**	–
w/o Quality Cue	$72.5_{\downarrow 6.4}$	$69.9_{\downarrow 6.3}$	$57.0_{\downarrow 4.8}$	-5.8
w/o Novelty Cue	$76.9_{\downarrow 2.0}$	$74.4_{\downarrow 1.8}$	$60.3_{\downarrow 1.5}$	-1.8
w/o Stability Cue	$\underline{77.8}_{\downarrow 1.1}$	$\underline{74.6}_{\downarrow 1.6}$	$\underline{60.6}_{\downarrow 1.2}$	<u>-1.3</u>

Novelty and stability cues also contribute to performance by preserving useful appearance changes and suppressing unstable updates. These results validate the effectiveness of the multi-cue design in EDTU.

V. Conclusion

In this paper, we revisit the RGB-T tracking task from a denoising perspective and propose a novel tracking framework D^3-Track, which consists of three core components: HCKD, TAGR, and EDTU. HCKD aligns RGB/TIR statistics at shallow layers, projects features to a shared subspace, and applies token-wise directional distillation to suppress semantic mismatch. TAGR assigns quality-aware, spatially adaptive fusion weights, while EDTU performs event-driven template updates with a four-template pool, mitigating drift without an explicit temporal encoder. Experiments on RGBT234, LasHeR, and VTUAV show consistent gains over strong baselines, especially under low illumination, clutter, and look-alike distractors.

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